

Project #5 guidelines for Python

If you decide to write your project using Python, I'll tell you where to implement the code in project #5 framework.

Implementation

1. Use your custom data preprocessing methods

- a. Complete ***Preprocessor class*** in *preprocessor.py*

Use your custom data preprocessing functions in project #1 as the methods in Preprocessor class.

```
4 class Preprocessor:
5     """ TODO """
6     def __init__(self, df):
7         # Initialize the preprocessor with a DataFrame
8         self.df = df
```

- b. Complete ***DataPreprocessing function*** in *main.py*

Use the custom methods in project #1 you implemented to complete dataPreprocessing function.

```
11 def dataPreprocessing():
12     """ TODO, use your own dataPreprocess function here. """
13
14     return
```

Note: You can either use your data preprocessing methods in project #1 or implement a new one for project #5.

2. Complete Naïve Bayes in *model.py*

- a. Implement necessary methods in *NaiveBayesClassifier*
(You don't have to follow the template to implement your NB)

```
23 # Naive Bayes Classifier
24 class NaiveBayesClassifier:
25     def __init__(self):
26         self.priors = {} # P(Class)
27         self.likelihoods = {} # P(Feature|Class)
28
29     def calculate_prior(self, y):
30         # TODO
31         pass
32
33     def calculate_likelihood(self, x, y):
34         # TODO
35         pass
36
37     def fit(self, x, y):
38         # TODO
39         pass
40
41     def predict(self, x):
42         # TODO
43         pass
44
45     def predict_proba(self, x):
46         # TODO
47         pass
```

3. Complete main function in *main.py*

- a. Build your Naïve Bayes and save the predict label as .csv file

```
18 def main():
19     root_path = r"yourtraining_data" # change the root path
20     train_X = os.path.join(root_path, "train_x.csv")
21     train_y = os.path.join(root_path, "train_y.csv")
22     test_X = os.path.join(root_path, "test_x.csv")
23
24     model = NaiveBayesClassifier()
25     model.fit(train_X, train_y)
26
27     # TODO
28     # build your NB model
29     # predict the output of the testing data
30     # remember to save the predict label as .csv file
31
```